

ABOUT

Security and Information Technology engineer. Since 2025 on a ZTNA SaaS platform (TypeScript/Node.js), autonomous and in direct contact with the CTO. Backend/platform: identity federation (Keycloak, Entra ID), RBAC, OpenAPI contracts consumed by the mobile, SDK and C++ network teams. Took the central console from prototype to a self-hostable solution, shipped to production.

PROFESSIONAL EXPERIENCE

Full-Stack Engineer (fixed-term contract) – Snowpack (ZTNA platform)

Paris

05/2026 - Present

- **3–6× lower API latency at p50, p90 and p99** (k6 load tests): backend run via ts-node without compilation, migrated to compiled ESM modules after resolving cyclic dependencies.
- **Resolved production incident:** CPU spikes (40–70% every 10 min) traced in Grafana to a gzip bomb; capped decompression size, restricted the API to JSON, refactored the cronjob system; CPU back to 10–20%.
- **Central console shipped as a self-hostable solution:** validated on Docker and Raspberry Pi; packaging: esbuild bundle, minification, single executable (Node SEA). Customer documentation and setup script.
- GitLab CI split into **two pipelines** (SaaS / customer build); every push runs the **TDD** suite (Vitest), including end-to-end tests on containerised MongoDB.
- **Fixed TOCTOU** on CRUD paths: application read-then-write replaced by MongoDB unique constraints; atomicity guaranteed at the database layer.
- **Stack:** TypeScript, Node.js, ESM, swc, esbuild, Node SEA, k6, Grafana, Docker, Vitest, GitLab CI.

INTERNSHIP: Full-Stack Developer – Snowpack (ZTNA platform)

Paris

02/2025 - 03/2026

- **Prototype to production in 2 months:** group management system designed end-to-end from a raw CTO requirement; polymorphic assignment of users/devices, automatic propagation of services and licences, deterministic recalculation on any membership or policy change.
- **Identity federation** (Keycloak, Microsoft Entra ID): strategy pattern per provider, accounts provisioned on first sign-in, no synchronisation job.
- **Generic typed task queue:** compile-time-typed event map, per-queue dispatcher, fire-and-forget and synchronous modes. Adopted by another developer unaided.
- **Central console** (API + web): single entry point for all Snowpack components (iOS/Android, SDK, network); OpenAPI contracts consumed by the mobile, SDK and C++ network teams.
- **Load time cut by 4× under 2G throttling** (React frontend): removed redundant refetching between contexts, caching and narrower queries.
- **Stack:** TypeScript, Node.js, TSOA / OpenAPI, React, MongoDB, TypeORM, Keycloak, Docker, GitLab CI.

PROJECTS

RFP proposal scorer – 24h hackathon - architecture lead

Frankfurt

09/2026

- **Multi-stage LLM scoring pipeline:** regex for non-semantic signals, LLM extraction with verbatim citations, three parallel scoring stages to bound context and cost per call; forced chain-of-thought traces weaknesses first, so every score derives from cited evidence.
- **Stack:** Python (FastAPI), React, Docker.
- **Github** : <https://github.com/UNtergas/SiviHack-2026-Team>

EDUCATION

INSA Centre Val de Loire – Engineering degree, Security and Information Technology

Bourges (2022-2026) · Selective Franco-Vietnamese integrated preparatory programme, Hué (2020-2022)

2020 - 2026

SKILLS

Languages: TypeScript, JavaScript, Python, SQL
Backend: Node.js, Express, TSOA, REST, OpenAPI / Swagger, FastAPI
Frontend: React, Next.js, Vite
Databases: MongoDB, TypeORM, SQL
Security & Identity: Authentication, RBAC, application security, Keycloak, Microsoft Entra ID
Testing: Vitest (TDD, end-to-end), k6 (load testing)
Build & DevOps: Docker, GitLab CI, ESM, swc, esbuild, Node SEA, Grafana
Practices: OOP, TDD